

Some Insights on Zeros of Fourier Transforms of Sets in $\mathbb{Z}_n^2$

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2025.09

Quick facts and notations

Fourier transform:

$$\widehat{1}_A(x) = \sum_{a \in A} e^{2\pi i \langle a, x \rangle / n}, \quad A \subseteq \mathbb{Z}_n^2$$

Zero set:

$$Z(\widehat{1}_A) = \{x \in \mathbb{Z}_n^2 : \widehat{1}_A(x) = 0\}$$

Difference set:

$$\Delta A = \{a - a' : a, a' \in A, a \neq a'\}$$

Quick facts and notations

$$(A, S) \text{ is a spectral pair on } \mathbb{Z}_n^2 \Leftrightarrow \begin{cases} |A| = |S| \\ \Delta S \subseteq Z(\hat{1}_A) \end{cases}$$

$$(A, B) \text{ is a tiling pair on } \mathbb{Z}_n^2 \Leftrightarrow \begin{cases} |A| \cdot |B| = n^2 \\ \Delta \mathbb{Z}_n^2 \subseteq Z(\hat{1}_A) \cup Z(\hat{1}_B) \end{cases}$$

Appetizer

Let H be a maximal cyclic subgroup in $\mathbb{Z}_n^2$ generated by h , then

$$\hat{1}_H(x) = \sum_{k=0}^{n-1} e^{2\pi i k \langle h, x \rangle / n} = \begin{cases} 0 & \text{if } \langle h, x \rangle \neq 0, \\ n & \text{if } \langle h, x \rangle = 0, \end{cases}$$

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Orthogonal set:

$$A^\perp = \{x \in \mathbb{Z}_n^2 : \langle a, x \rangle = 0, \forall a \in A\}.$$

Poisson summation formula (maximal cyclic subgroups)

If H is a maximal cyclic subgroup, then

$$Z(\hat{1}_H) = \mathbb{Z}_n^2 \setminus H^\perp$$

More precisely

$$\hat{1}_H = |H| \cdot 1_{H^\perp}$$

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Spoiler: This actually holds for all subgroups.

Examples

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Basic observation

Given some A , let's consider whether $x = (k, 0)$ is in $Z(\hat{1}_A)$:

$$\hat{1}_A(x) = \sum_{(a_1, a_2) \in A} e^{2\pi i(a_1 \cdot k + a_2 \cdot 0)/n} = \sum_{(a_1, a_2) \in A} e^{2\pi i a_1 \cdot k/n}$$

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(1) If A_1 is the multiset obtained by applying $(a_1, a_2) \mapsto (a_1, 0)$ on A , then $\hat{1}_A(x) = \hat{1}_{A_1}(x)$ for $x = (k, 0)$.

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(2) We may view $\sum_{(a_1, a_2) \in A} e^{2\pi i a_1 \cdot k/n}$ as the univariate polynomial

$$p(z) = \sum_{(a_1, a_2) \in A} z^{a_1}$$

evaluated at $z = e^{2\pi i k/n}$.

The idea

Given some A and x , to determine whether $x \in Z(\hat{1}_A)$, we find some x', y' so that $x \in \langle x' \rangle$, $x' \perp y'$, and x', y' generates $\mathbb{Z}_n^2$.

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x', y' is a pair of “orthogonal Euclidean basis”, and if $x' = (x_1, x_2)$, then $y' = (-x_2, x_1)$.

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Write every element $a \in A$ into linear combinations of x', y' , then repeating the same procedure we obtain

(1) $\hat{1}_A(x)$ depends only on the projection of A into $\langle x' \rangle$;

(2) $\hat{1}_A(x)$ can be viewed as a univariate polynomial evaluated at $z = e^{2\pi i k/n}$ where k is the value so that $x = kx'$.

Facts: disjointness of difference sets

Fact 1:

$$A + B = A \oplus B \quad \Leftrightarrow \quad \Delta A \cap \Delta B = \emptyset$$

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Define an equivalence relation in finite Abelian groups: $h \sim h'$ if h, h' generate the same cyclic subgroup.

Fact 2: Let E be an arbitrary equivalence class under $\sim$, then $\Delta A \cap E$ and $\Delta B \cap E$ can not be both non-empty.

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Reason(dilation trick): If $\gcd(m, |A|) = 1$ and (A, B) is a tiling pair, then (mA, B) is still a tiling pair.

Equivalence classes

Yellow	Teal	Olive	Light Blue
Green	Pink	Blue	Pink
Yellow	Light Blue	Olive	Teal
White	Pink	Red	Pink

Facts: a representation of $\mathbb{Z}_{p^k}$

Expand every element of $\mathbb{Z}_{p^k}$ in base p , then each element can be represented by a k -digit base p number.

Fact: Subgroups have common trailing 0s:

A subgroup of size p^t consists of all numbers whose last $k - t$ digits are 0, and its generators are of the form:

$$d_1 \dots d_{t-1} \times \underbrace{0 \dots 0}_{k-t \text{ digits}},$$

with $x \neq 0$ and $d_1, \dots, d_{t-1}$ arbitrary.

Quick facts and notations

Let

$$C' = \{\underbrace{0 \dots 0}_{t-1 \text{ digits}} \times \underbrace{0 \dots 0}_{k-t \text{ digits}} : x \in \mathbb{Z}_p\},$$

then $|C'| = p$, and both $C', \Delta C'$ belong to the same equivalence class.

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Let E be an equivalence class in $\mathbb{Z}_{p^k}^2$, and $h \in E$.

E generates a cyclic group of some prime power order, hence if

$$E' = \{0, h, \dots, (p-1)h\},$$

then $|E'| = p$, and both $E', \Delta E' \subseteq E$.

Quick facts

All elements in a given equivalence class are simultaneously annihilated.

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Each equivalence class E contains an size p subset E' whose difference set is still in the class.

We will construct sets of form $E'_1 \oplus E'_2 \oplus \dots \oplus E'_m$ as spectra for tilings sets or tiling complements for spectral sets.

Some immediate consequences

Lemma

If $h \in Z(\hat{1}_A)$, then $\hat{1}_A$ annihilates the whole equivalence class of h .

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Theorem

Tiling sets and spectral sets coincides in $\mathbb{Z}_p^2$.

$T \Rightarrow S: |A| = p \text{ in } \mathbb{Z}_{p^k}^2$

Let (A, B) be a tiling pair in $\mathbb{Z}_{p^k}^2$ with $|A| = p$.

$Z(\hat{1}_A) \neq \emptyset$ must hold, otherwise $Z(\hat{1}_B)$ would contain the whole $\Delta \mathbb{Z}_{p^k}^2$.

There is some equivalence class E with $E \subseteq Z(\hat{1}_A)$, and E' is a spectrum of A .

$$S \Rightarrow T: |A| \geq p^{2k-1} \text{ in } \mathbb{Z}_{p^k}^2$$

Let (A, S) be a spectral pair in $\mathbb{Z}_{p^k}^2$ with $|A| \geq p^{2k-1}$.

There must be some equivalence class E that is disjoint with ΔA , otherwise $Z(\hat{1}_S)$ would contain the whole $\Delta \mathbb{Z}_{p^k}^2$.

$\Delta E' \cap \Delta A \subseteq E \cap \Delta A = \emptyset$, thus E' is a tiling complement of A .

Tiling: $|A| = p^{2k-1}$ in $\mathbb{Z}_{p^k}^2$

Let (A, B) be a tiling pair in $\mathbb{Z}_{p^k}^2$ with $|A| = p^{2k-1}$.

There must be some equivalence class E that is disjoint with ΔA , otherwise there is no place for ΔB .

There is an S with $|S| = p^{2k-1}$ and $Z(\hat{1}_S) = \Delta \mathbb{Z}_{p^k}^2 \setminus E$.

S is essentially of form $H^\perp \oplus X'$, where H is a maximal cyclic subgroup that contains E , while X is the equivalence class of the subgroup of order $|H|/|E|$ in H . (that S takes such a form will be more evident in the symplectic setting)

The symplectic settings

Symplectic form (anti-symmetric, bilinear):

$$\langle x, y \rangle_s = x_1 y_2 - x_2 y_1 = \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle,$$

where $x = (x_1, x_2), y = (y_1, y_2)$.

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Remark: The Fourier transform incurs a 90 degree rotation on the time-frequency plane, the rotation matrix is the symplectic matrix.

Appetizer revisited

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Symplectic orthogonal sets:

$$H^{\perp_s} = \{x \in \mathbb{Z}_n^2 : \langle h, x \rangle_s = 0\}.$$

Examples

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VS

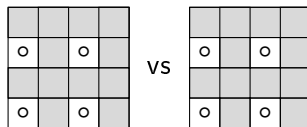
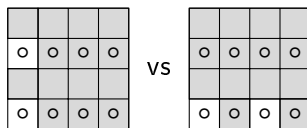
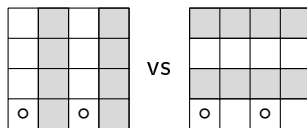
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VS

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More examples



Characterizing orthogonal sets

$A^\perp$ and $A^{\perp_s}$ are always subgroups. If $ab = n$, then

$$(a\mathbb{Z}_b \times \mathbb{Z}_n)^\perp = b\mathbb{Z}_a \times \{0\}, \quad (a\mathbb{Z}_b \times \mathbb{Z}_n)^{\perp_s} = \{0\} \times b\mathbb{Z}_a.$$

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Fact 1: A set complements a subgroup H in $\mathbb{Z}_n^2$ if and only if its size is $n^2/|H|$ and it annihilates $H^\perp$ (in the Euclidean setting) or $H^{\perp_s}$ (in the symplectic setting).

Fact 2: H is an order n subgroup in $\mathbb{Z}_n^2$ if and only if $H = H^{\perp_s}$.

Fact 3: Poisson summation formula: If H is a subgroup, then

$$\widehat{1_H^s} = |H| \cdot 1_{H^{\perp_s}}$$

Strategy and motivation revisited

To determine whether $h \in Z(\hat{1}_A)$ or $Z(\hat{1}_A^s)$, first we view the group as generated by linear combinations of x', y' , where $h \perp y'$ or $h \perp_s y'$.

Strategy and motivation revisited

To determine whether $h \in Z(\hat{1}_A)$ or $Z(\hat{1}_A^s)$, first we view the group as generated by linear combinations of x', y' , where $h \perp y'$ or $h \perp_s y'$.

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Next we project A to $\langle x' \rangle$, and view the result as a univariate polynomial evaluated at some point, the polynomial can then be factorized.

Direct computation is often complicated, ideally we shall find a map that preserves $\langle \cdot, \cdot \rangle$ or $\langle \cdot, \cdot \rangle_s$ (preserves tilingness and spectrality), and can send orthogonal basis to standard basis.

The Euclidean case

Let us inspect whether this is possible for $\langle \cdot, \cdot \rangle$:

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Suppose x, y generate $\mathbb{Z}_n^2$ and $x \perp y$:

(1) If $\langle Ua, Ub \rangle = \langle a, b \rangle$ holds for all a, b , then $UU^* = I$.

(2) If $Ux = (1, 0)^T$, $Uy = (0, 1)^T$, then $U \begin{pmatrix} x & y \end{pmatrix} = I$, hence $U = \begin{pmatrix} x^T \\ y^T \end{pmatrix}$
and $x_1 \cdot x_1 + x_2 \cdot x_2 = y_1 \cdot y_1 + y_2 \cdot y_2 = 1$.

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and $x_1 \cdot x_1 + x_2 \cdot x_2 = y_1 \cdot y_1 + y_2 \cdot y_2 = 1$.

Conclusion: NOT always possible in $\mathbb{Z}_n^2$.

Example

No such U exists for orthogonal pairs from $\{(0,0), (1,1), (2,2), (3,3)\}$ and $\{(0,0), (1,3), (2,2), (3,1)\}$:

			x
	x		

	y		
			y

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For $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, the candidate U should be $2^{-1} \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$, but 2 is not multiplicatively invertible in $\mathbb{Z}_4$.

Symplectic bases

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Fact: Symplectic form is preserved on change of bases.

If $x = x_1 h + x_2 h', y = y_1 h + y_2 h'$, then

$$\begin{aligned}\langle x, y \rangle_s &= x_1 y_1 \langle h, h \rangle_s + x_1 y_2 \langle h, h' \rangle_s + x_2 y_1 \langle h', h \rangle_s + x_2 y_2 \langle h', h' \rangle_s \\ &= x_1 y_2 - x_2 y_1\end{aligned}$$

Symplectomorphisms

Symplectomorphism: change of symplectic bases

Fact: On $\mathbb{Z}_n^2$ they are 2×2 matrices with $\det = 1$.

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(1) Unit determinant matrices preserve the symplectic form:

If $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, then $\langle x, y \rangle_S = \langle x, Sy \rangle$, and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies $A^*SA = S$ if and only if $ad - bc = 1$.

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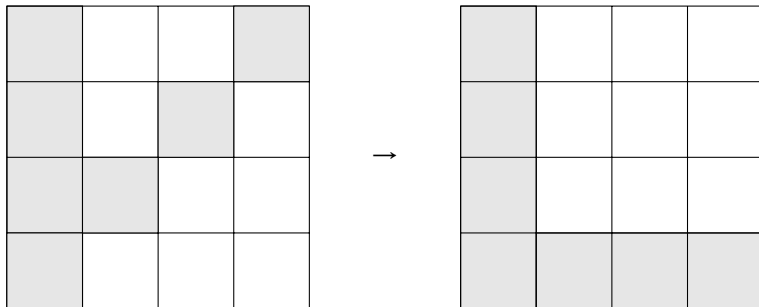
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(2) Matrices formed by symplectic basis have unit determinant:

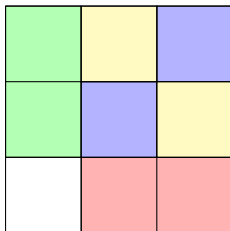
$$\langle x, y \rangle_S = x_1y_2 - x_2y_1 = \det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$$

Example



Any pair of generating maximal cyclic groups would work, they don't have to be orthogonal in the Euclidean sense.

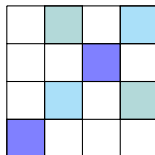
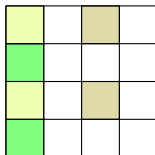
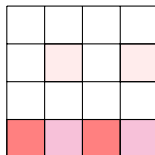
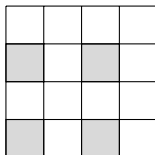
Structures of $\mathbb{Z}_p^2$



The group dissects into maximal cyclic subgroups that mutually intersect trivially.

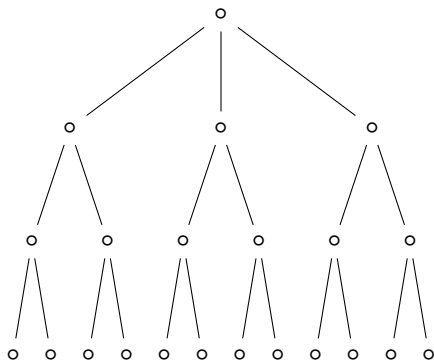
There are $p+1$ such subgroups, each subgroup forms a line.

Structures of $\mathbb{Z}_{p^2}^2$



For every order p cyclic group H , there are precisely p maximal cyclic groups that intersect at H .

Structures of $\mathbb{Z}_{p^k}^2$



• • •

Each path from the root is a chain of cyclic subgroups by inclusion.

$$T \Rightarrow S \text{ in } \mathbb{Z}_{p^2}^2$$

Theorem

Tiling sets are also spectral in $\mathbb{Z}_{p^2}^2$.

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Suffices to consider tiling sets of size p^2 . Let (A, B) be such a pair.

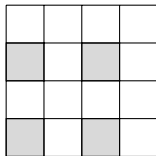
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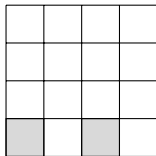
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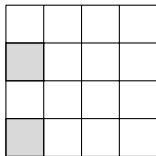
Denote and distinguish order p subgroups by their “slopes”:



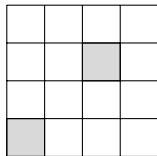
K



K_0



K_∞



K_1

$$T \Rightarrow S \text{ in } \mathbb{Z}_{p^2}^2$$

Let $M = \{i : \Delta K_i \subseteq Z(\hat{1}_A^s)\}$ and $M' = \{i : \Delta K_i \cap Z(\hat{1}_A^s) = \emptyset\}$

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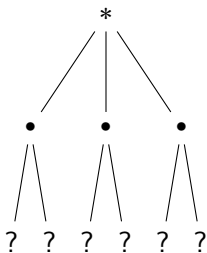
(3) If $0 < |M| < p + 1$, then depending on the status of leaf nodes:

(3.1) A annihilates a maximal path;

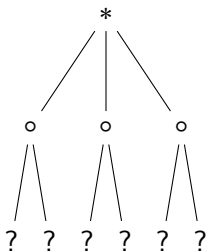
(3.2) A annihilates a full sub-branch of some $m' \in M'$;

(3.3) No other cases possible.

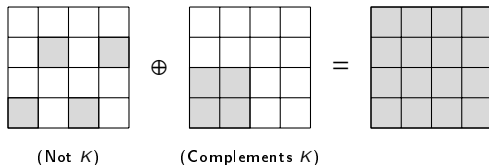
T2S case (1)



T2S case (2)



That B complementing K does not mean A is K , an example:



A counting lemma

Lemma

If $x \in Z(\hat{1}_A^s)$, and $\text{ord}(x) = p$ is a prime power, then

$$|A \cap \langle px \rangle^{\perp_s}| = p \cdot |A \cap \langle x \rangle^{\perp_s}|$$

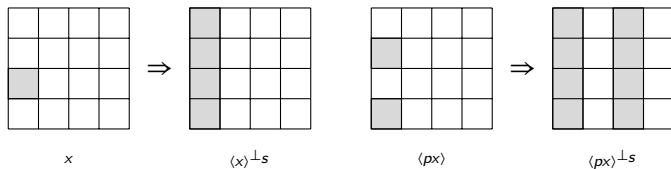
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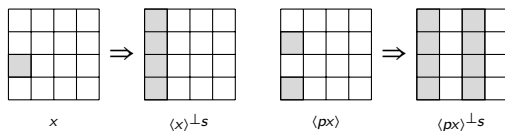
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For illustration purpose, let's say $x = (0, 1)$:



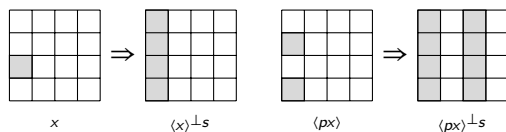
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Let $\omega = e^{2\pi i/p^2}$, $P(z) = \sum_{(a_1, a_2) \in A} z^{a_1}$, then

$$\hat{1}_A^s(x) = \sum_{(a_1, a_2) \in A} e^{2\pi i(a_1 \cdot 1 - a_2 \cdot 0)/p^2} = \sum_{(a_1, a_2) \in A} \omega^{a_1} = P(\omega).$$

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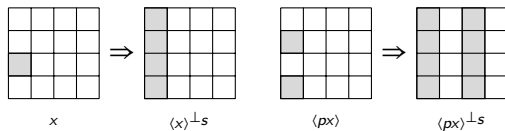
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If $x \in Z(\hat{1}_A^s)$, then $P(\omega) = 0$, which means $P(z)$ is divisible by the p^2 -th cyclotomic polynomial, hence

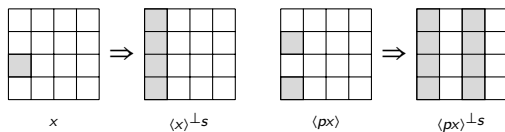
$$P(\omega) = (1 + \omega^p + \dots + \omega^{p(p-1)})(c_0 + c_1\omega + \dots + c_{p-1}\omega^{p-1}).$$

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$$\begin{aligned}\hat{1}_A^s(x) &= \sum_{a \in A} \omega^{\langle a, x \rangle_s} = \sum_{(a_1, a_2) \in A} \omega^{a_1} \\ &= (1 + \omega^p + \dots + \omega^{p(p-1)})(c_0 + c_1\omega + \dots + c_{p-1}\omega^{p-1}).\end{aligned}$$

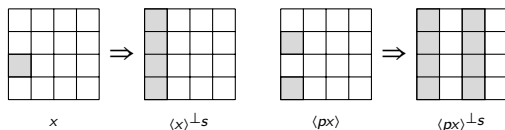
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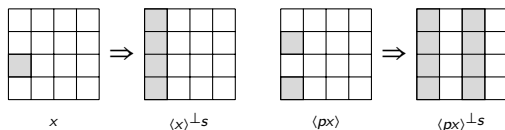


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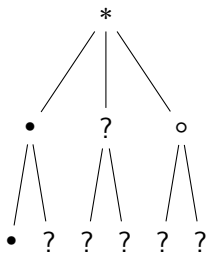
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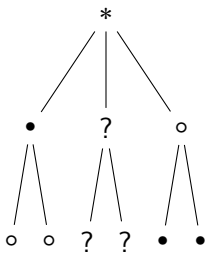
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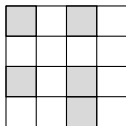
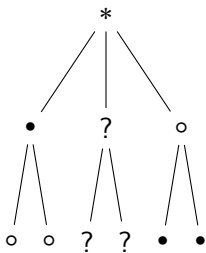
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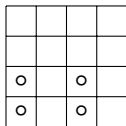
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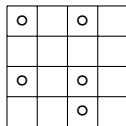
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(in $Z(\hat{1}_A^S)$)

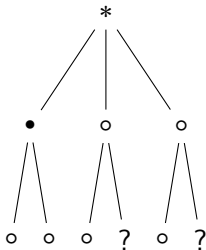


$S = \{(0,0), (0,1)\} \oplus \{(0,0), (2,0)\}$

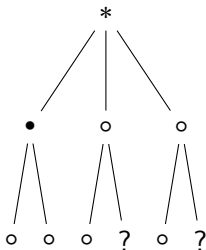


$\Delta S = \Delta E' \oplus \Delta K_0$

T2S case (3.3)

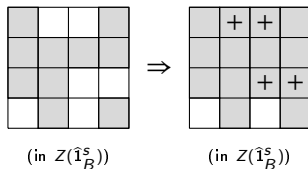


T2S case (3.3)



B has to annihilate the white nodes, which implies that

B annihilates $\mathbb{Z}_{p^2} \setminus K$ and all ΔK_i for i in M' , it has too many zeros!



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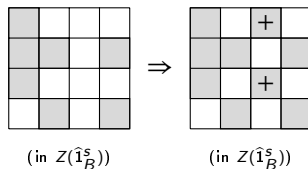
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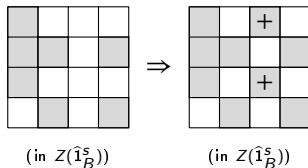
which means $|A| \geq n^2 / |H^\perp s| = |H| > |A|$.

Excessive zeros



$\Delta(\{0\} \times \mathbb{Z}_{p^2}) \subseteq Z(\hat{1}_B^s)$ implies $B = \{(0, b_0), \dots, (p^2 - 1, b_{p^2-1})\}$.

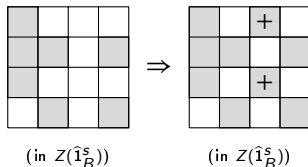
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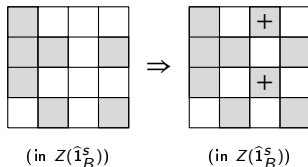


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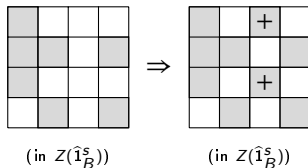
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The aim is to produce a B so that $\Delta A \cap \Delta B = \emptyset$, and $|B| = p^2$ or p^3 respectively.

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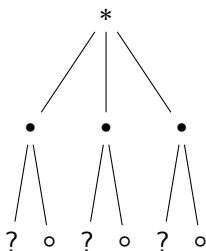
If $m' \in M'$, and $H' = E \cup K_{m'}$ is a maximal cyclic subgroup, then $E \cap \Delta A \neq \emptyset$ (since $\Delta A \cap \Delta H$ has to be non-empty).

(2.1) $|M'| = 0$: S is a tiling complement of K , mutual annihilation.

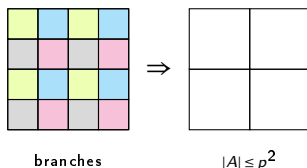
(2.2) $|M'| > 0$, $\exists m \in M$ with empty leaf: complements some $K_i \oplus E'$.

(2.3) Else: impossible.

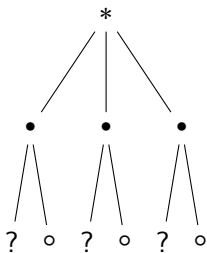
$S2T : p^2 \leq |A| < p^3$ case (2.1)



For each K_i , there shall be some maximal cyclic subgroup H_i so that $\Delta A \cap \Delta H_i \subseteq K_i$, otherwise $\Delta K_i^{\perp_S} \subseteq Z(\hat{1}_S^S)$ (S annihilates a larger subgroup). This implies that $|A| = p^2$.

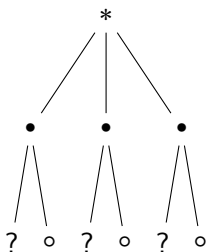


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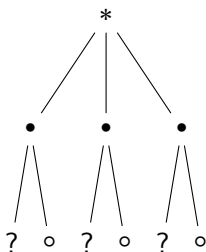
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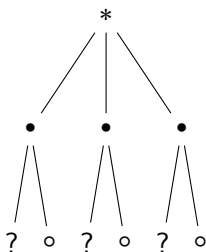


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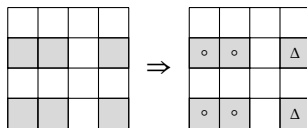
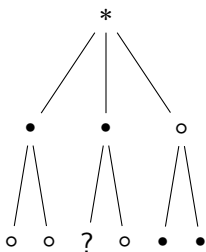
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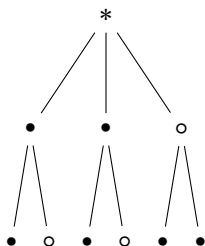
$S2T : p^2 \leq |A| < p^3$ case (2.2)



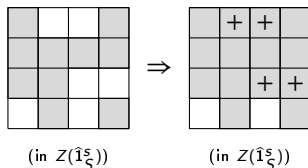
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$S2T : p^2 \leq |A| < p^3$ case (2.3)



S annihilates $\mathbb{Z}_{p^2} \setminus K$ and all ΔK_i for i in M , it has too many zeros!



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The branch of every $m \in M$ has at least one empty leaf node.

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(1) ΔA intersects each order p^2 subgroup non-trivially:

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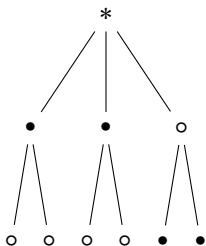
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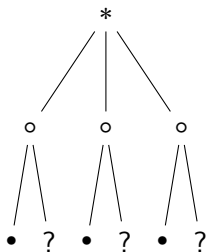
(3) $\Delta A \cap \Delta H = \emptyset$ for some maximal cyclic subgroup H .

Reduce to $\mathbb{Z}_{p^2} \times \mathbb{Z}_p$.

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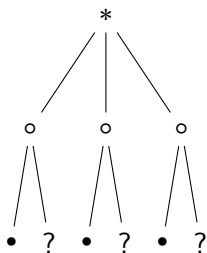


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At least one leaf in each branch for $\Delta(A \oplus K)$ to intersect all $\Delta E'$.

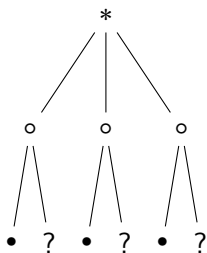
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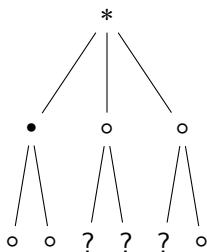


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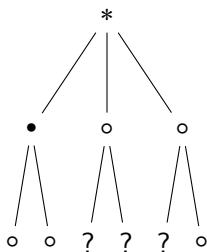
Sum over i : $|S| + p|S \cap K| = p^2 + p|S \cap K| + p|S \cap E_i| \Rightarrow |S| \geq p^2$.

$S2T : p \leq |A| < p^2$ case (3)



Let $a_i = |A \cap (K_i^{\perp_s} \setminus K)|$, $b = |A \cap K|$, $a'_i = |S \cap (K_i^{\perp_s} \setminus K)|$, $b' = |S \cap K|$.

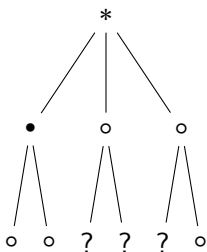
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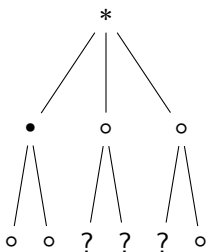


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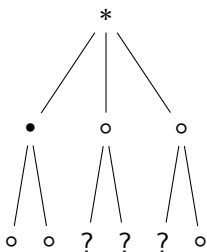
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Structures and results on $\mathbb{Z}_{p^2}^2$ are briefed by a case discussion along its tree structure.

